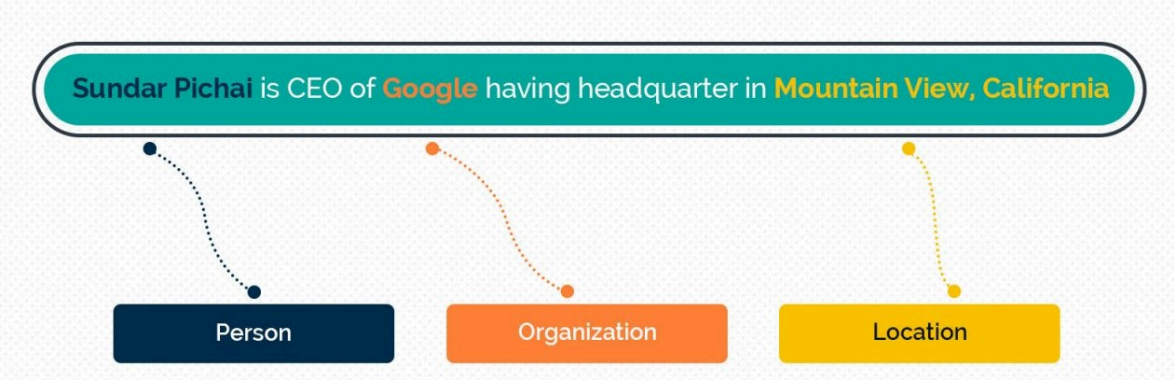
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Name Entity Recognition (NER)

Name Entity Recognition (NER) is a fundamental task in NLP that focuses on identifying and classifying entities such as names of people, organizations, locations, dates, and more within a given text. NER plays a crucial role in various NLP applications, including information retrieval, question answering, and sentiment analysis.



At its core, it is just a two-step process:

1. Detecting the entities from the text
2. Classifying them into different categories

NER using spaCy

SpaCy offers robust support for Named Entity Recognition (NER). It can recognize a variety of named entities, such as persons, organizations, dates, locations, and more. Here's more information about using spaCy for NER:

TYPE	DESCRIPTION
PERSON	People, including fictional.
NORP	Nationalities or religious or political groups.
FAC	Buildings, airports, highways, bridge, etc.
ORG	Companies, agencies, institutions etc.
GPE	Countries, cities, states.
LOC	Non-GPE locations, mountain ranges, bodies of water.
PRODUCT	Objects, vehicles, foods, etc. (Not services)
EVENT	Named hurricanes, battles, wars, sports events, etc.
WORK_OF_ART	Titles of books, songs, etc.
LAW	Named documents made into laws.
LANGUAGE	Any named language.
DATE	Absolute or relative dates or periods.
TIME	Times smaller than a day.
PERCENT	Percentage, including "%".
MONEY	Monetary values, including unit.
QUANTITY	Measurements, as of weight or distance.
ORDINAL	"first" , "second" , etc.
CARDINAL	Nuerals that do not fall under another type.

When performing Named Entity Recognition (NER) using spaCy or other NLP tools, you'll encounter various entity types, and they are represented as labels. Here are some common entity types and their labels:

Entity Type	Label	Description	Examples
Person	`PERSON`	Represents names of people.	*John Smith, Barack Obama*
Organization	`ORG`	Names of companies, institutions, agencies, or other organized entities.	*Microsoft, United Nations*
Location	`GPE`, `LOC`	Geographical locations such as cities, countries, or landmarks.	*New York, Eiffel Tower*
Date	`DATE`	Calendar dates, time intervals, or relative dates.	*January 1, 2023, tomorrow*
Time	`TIME`	Clock times or time intervals.	*2:30 PM, 5 minutes*
Money	`MONEY`	Monetary values, including currencies and amounts.	*$100, €50*
Percent	`PERCENT`	Percentage values.	*50%, 10 percent*
Product	`PRODUCT`	Names of products or goods.	*iPhone, Coca-Cola*
Event	`EVENT`	Names of events or occurrences.	*World Cup, New Year's Eve*
Law	`LAW`	Legal references or statutes.	*Copyright Act, First Amendment*
Language	`LANGUAGE`	Names of languages.	*English, Spanish*
Nationality/Religion/Political Group	`NORP`	Nationalities, religious groups, or political affiliations.	*American, Buddhist*

In this example, we will use spaCy to perform NER on some sample text. Make sure you have spaCy installed; if not, you can install it using : `pip install spacy`

In [1]:
```
%pip -q install "spacy>=3.8,<3.9"
import importlib.util
if importlib.util.find_spec("en_core_web_sm") is None:
    %pip -q install https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any.whl
```

```
Note: you may need to restart the kernel to use updated packages.
```

In [2]:
```
# Import the spaCy library and load the English language model
import spacy
nlp = spacy.load("en_core_web_sm")

# Sample text for NER
text = "Apple Inc. was founded by Steve Jobs in Cupertino, California. It is known for its iPhone and Mac products."
doc = nlp(text)

print(f"Entities recognized in text : {doc.ents}") #Return a tuple of entities recognized in text

# Extract and print named entities
for entity in doc.ents:
    print(f"Entity: {entity.text}, Label: {entity.label_}")
```

```
Entities recognized in text : (Apple Inc., Steve Jobs, Cupertino, California, Mac)
Entity: Apple Inc., Label: ORG
Entity: Steve Jobs, Label: PERSON
Entity: Cupertino, Label: GPE
Entity: California, Label: GPE
Entity: Mac, Label: ORG
```

In [3]:

```
# Iterate through recognized entities and print their details
for ent in doc.ents:
    print(f"Entity: {ent.text}")
    print(f"Label: {ent.label_}")
    print(f"Start Position: {ent.start}")
    print(f"End Position: {ent.end}")
    print(f"Entity Type: {spacy.explain(ent.label_)}")
    print("")
```

```
Entity: Apple Inc.
Label: ORG
Start Position: 0
End Position: 2
Entity Type: Companies, agencies, institutions, etc.

Entity: Steve Jobs
Label: PERSON
Start Position: 5
End Position: 7
Entity Type: People, including fictional

Entity: Cupertino
Label: GPE
Start Position: 8
End Position: 9
Entity Type: Countries, cities, states

Entity: California
Label: GPE
Start Position: 10
End Position: 11
Entity Type: Countries, cities, states

Entity: Mac
Label: ORG
Start Position: 19
End Position: 20
Entity Type: Companies, agencies, institutions, etc.
```

Create a custom Named Entity Model

In spaCy, you can add custom entities that have not been identified in the pre-trained models by training your own Named Entity Recognition (NER) model. This involves providing labeled examples of your custom entities and then fine-tuning the model.

Here is an example of how to train a custom NER model in spaCy:
https://spacy.io/usage/training#ner

In [4]:

```
from spacy.training.iob_utils import offsets_to_biluo_tags

# The text to annotate
text = "Apple Inc. was founded by Steve Jobs in Cupertino, California. It is known for its iPhone and Mac products."

# --- Add custom entities annotations manually ---
annotations = [
    (text.find("Apple Inc."), text.find("Apple Inc.") + len("Apple Inc."), "ORG"),
    (text.find("Steve Jobs"), text.find("Steve Jobs") + len("Steve Jobs"), "PERSON"),
    (text.find("Cupertino, California"), text.find("Cupertino, California") + len("Cupertino, California"), "GPE")]

# --- Step 2: Check entity alignment ---
doc = nlp.make_doc(text) #splitted to tokens
tags = offsets_to_biluo_tags(doc, annotations) #converts annotations into BILUO tags (Begin, Inside, Last, Unit, Outside).
print("Alignment Tags:", tags[:10])
```

```
Alignment Tags: ['B-ORG', 'L-ORG', 'O', 'O', 'O', 'B-PERSON', 'L-PERSON', 'O', 'B-GPE', 'I-GPE']
```

In [5]:

```
from spacy.training import Example
from spacy.util import fix_random_seed

fix_random_seed(42)
custom_nlp = spacy.load("en_core_web_sm")
example = Example.from_dict(custom_nlp.make_doc(text), {"entities": annotations})
with custom_nlp.select_pipes(enable=["tok2vec", "ner"]):
    optimizer = custom_nlp.resume_training()
    for _ in range(20):
        custom_nlp.update([example], sgd=optimizer, drop=0.2)
custom_nlp.to_disk("custom_ner_model")
```

Visualization

In [6]:

```
# Load and use the custom model for entity recognition
custom_nlp = spacy.load("custom_ner_model")
doc2 = custom_nlp(text)

# Entity visualization
from spacy import displacy
displacy.render(doc2, style="ent", jupyter=True)
```

Apple Inc. **ORG** was founded by Steve Jobs **PERSON** in Cupertino, California **GPE** . It is known for its iPhone and Mac products.

Task 1

1. Ask a user to enter a sentence, and identify the named entities in the sentence.

2. Print the details of each entity in the sentence provided by the user.

In [7]:

```
sentence = input("Enter a sentence: ")
sentence_doc = nlp(sentence)
print("Entities:", sentence_doc.ents)
for ent in sentence_doc.ents:
    print(f"Entity: {ent.text}")
    print(f"Label: {ent.label_}")
    print(f"Start Position: {ent.start}")
    print(f"End Position: {ent.end}")
    print(f"Entity Type: {spacy.explain(ent.label_)}")
    print()
```

```
Enter a sentence: Steve Jobs founded Apple in California.
Entities: (Steve Jobs, Apple, California)
Entity: Steve Jobs
Label: PERSON
Start Position: 0
End Position: 2
Entity Type: People, including fictional

Entity: Apple
Label: ORG
Start Position: 3
End Position: 4
Entity Type: Companies, agencies, institutions, etc.

Entity: California
Label: GPE
Start Position: 5
End Position: 6
Entity Type: Countries, cities, states
```

Task 2

Use the paragraph below to answer the following questions:

“This agreement is made and entered into by and between ‘Abc & Co.’ and ‘Bcd LLC’ for term 1 year starting from April 1, 2020, hereinafter collectively referred to as the Parties. Samuel P. Jackson in the place (New York) and on the date written below, with the following terms and conditions.”

1. Find any entity that has not been identified and add it.
2. Find the count of ‘PERSON’ and ‘ORG’ in the paragraph above.
3. Visualize the entities in the paragraph.

In [8]:

```
from spacy.tokens import Span
from collections import Counter

paragraph = '“This agreement is made and entered into by and between ‘Abc & Co.’  and ‘Bcd LLC’ for term 1 year starting
from April 1, 2020,  hereinafter collectively referred to as the Parties. Samuel P. Jackson in the place (New York) and on
the date written below, with the following terms and conditions.”'
paragraph_doc = nlp(paragraph)
print("Original entities:", [(ent.text, ent.label_) for ent in paragraph_doc.ents])

expected_entities = [
    ("Abc & Co.", "ORG"),
    ("Bcd LLC", "ORG"),
    ("1 year", "DATE"),
    ("April 1, 2020", "DATE"),
    ("Samuel P. Jackson", "PERSON"),
    ("New York", "GPE"),
]
for entity_text, label in expected_entities:
    start = paragraph.index(entity_text)
    span = paragraph_doc.char_span(start, start + len(entity_text), label=label)
    assert span is not None
    if not any(ent.start == span.start and ent.end == span.end
               and ent.label_ == label for ent in paragraph_doc.ents):
        retained = [ent for ent in paragraph_doc.ents
                    if ent.end <= span.start or ent.start >= span.end]
        paragraph_doc.ents = sorted(retained + [span], key=lambda ent: ent.start)
        print("Added:", span.text, span.label_)

counts = Counter(ent.label_ for ent in paragraph_doc.ents)
print("PERSON:", counts["PERSON"])
print("ORG:", counts["ORG"])
displacy.render(paragraph_doc, style="ent", jupyter=True)
```

```
Original entities: [('between ‘Abc & Co.', 'ORG'), ('1 year', 'DATE'), ('April 1, 2020', 'DATE'), ('Samuel P. Jackson',
'PERSON'), ('New York', 'GPE')]
Added: Abc & Co. ORG
Added: Bcd LLC ORG
PERSON: 1
ORG: 2
```

“This agreement is made and entered into by and between ‘ Abc & Co. ORG ’ and ‘ Bcd LLC ORG ’ for term 1 year DATE starting from April 1, 2020 DATE , hereinafter collectively referred to as the Parties. Samuel P. Jackson PERSON in the place (New York GPE) and on the date written below, with the following terms and conditions.”

Task 3: Training a Custom NER Model

text = "Microsoft was founded by Bill Gates and Paul Allen in Albuquerque, New Mexico. It is known for its Windows operating system and Office software products."

1. Create a training example using `Example.from_dict()` with the provided text and annotations.
2. Train the NER model for **20 iterations**
3. Save the trained model in a folder named `custom_ner_model`.
4. Load the saved model and use it to identify named entities in a sentence.

In [9]:

```
from spacy.training import Example
from spacy.util import fix_random_seed

text = "Microsoft was founded by Bill Gates and Paul Allen in Albuquerque, New Mexico. It is known for its Windows
operating system and Office software products."
entity_labels = [
    ("Microsoft", "ORG"),
    ("Bill Gates", "PERSON"),
    ("Paul Allen", "PERSON"),
    ("Albuquerque", "GPE"),
    ("New Mexico", "GPE"),
    ("Windows", "PRODUCT"),
    ("Office", "PRODUCT"),
]
annotations = {"entities": [
    (text.index(entity), text.index(entity) + len(entity), label)
    for entity, label in entity_labels
]}
fix_random_seed(42)
training_nlp = spacy.load("en_core_web_sm")
ner = training_nlp.get_pipe("ner")
for _, label in entity_labels:
    ner.add_label(label)
example = Example.from_dict(training_nlp.make_doc(text), annotations)
with training_nlp.select_pipes(enable=["tok2vec", "ner"]):
    optimizer = training_nlp.resume_training()
    for iteration in range(20):
        losses = {}
        training_nlp.update([example], sgd=optimizer, drop=0.2, losses=losses)
        print(f"Iteration {iteration + 1}: {losses['ner']:.4f}")
training_nlp.to_disk("custom_ner_model")

custom_nlp = spacy.load("custom_ner_model")
test_sentence = "Bill Gates and Paul Allen founded Microsoft in Albuquerque, New Mexico."
print(test_sentence)
for ent in custom_nlp(test_sentence).ents:
    print(f"Entity: {ent.text}, Label: {ent.label_}")
```

```
Iteration 1: 2.9680
Iteration 2: 1.6611
Iteration 3: 0.9807
Iteration 4: 0.0052
Iteration 5: 0.0003
Iteration 6: 0.0001
Iteration 7: 0.0000
Iteration 8: 0.0002
Iteration 9: 0.0000
Iteration 10: 0.0013
Iteration 11: 0.0010
Iteration 12: 0.0000
Iteration 13: 0.0000
Iteration 14: 0.0000
Iteration 15: 0.0000
Iteration 16: 0.0000
Iteration 17: 0.0000
Iteration 18: 0.0000
Iteration 19: 0.0000
Iteration 20: 0.0000
```

```
Bill Gates and Paul Allen founded Microsoft in Albuquerque, New Mexico.
Entity: Bill Gates, Label: PERSON
Entity: Paul Allen, Label: PERSON
Entity: Microsoft, Label: ORG
Entity: Albuquerque, Label: GPE
Entity: New Mexico, Label: GPE
```